Technical Supplement: detailed domain diagnostics

Companion to *Controlled Instrument-Relevance Sensitivity in High-Dimensional IVQR: A Monte Carlo Study*. Tables A.2–A.5 retain their original captions, entries and numbering; Table A.1 and the essential interpretation remain in inline Appendix A of the proposed manuscript. These are the existing 100-replication domain diagnostics, not new results or the main 500-replication experiment. The recorded Stage 1 singular-design error remains an incomplete profile, not acceptance; its affected expanded-domain cell has 99 usable replications. Boundary acceptance concerns either endpoint and full-grid acceptance concerns all evaluated points; neither establishes continuous or global geometry. Med. and Mean denote replication medians and means; R is intended replications, Succ. complete profiles, Fail incomplete profiles and Stage 1/2 newly recorded evaluation errors.

Table A.2: Grid-domain diagnostic: added accepted measure

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	2.000	1.974	2.000	1.844
1000	0.10	0.25	Full-GMM	2.500	2.470	2.200	2.263
1000	0.10	0.25	DML-IVQR	2.150	2.433	2.000	2.205
1000	0.10	0.10	Oracle-GMM	4.000	3.636	4.000	3.621
1000	0.10	0.10	Full-GMM	3.850	3.610	3.875	3.580
1000	0.10	0.10	DML-IVQR	3.700	3.374	3.450	3.120
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.275	0.933	0.000	0.806
1000	0.25	0.25	Full-GMM	1.025	1.251	0.600	0.978
1000	0.25	0.25	DML-IVQR	1.750	1.500	2.000	1.582
1000	0.25	0.10	Oracle-GMM	3.900	3.207	3.900	3.174
1000	0.25	0.10	Full-GMM	3.800	3.296	3.800	3.192
1000	0.25	0.10	DML-IVQR	3.950	3.275	3.625	3.058
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.000	0.722	0.000	0.547
1000	0.50	0.25	Full-GMM	0.100	0.803	0.000	0.630
1000	0.50	0.25	DML-IVQR	0.250	0.895	0.400	0.921
1000	0.50	0.10	Oracle-GMM	3.900	3.119	4.000	3.047
1000	0.50	0.10	Full-GMM	3.900	3.182	3.900	3.099
1000	0.50	0.10	DML-IVQR	3.850	3.134	2.850	2.914
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	1.325	1.317	0.400	1.045
1000	0.75	0.25	Full-GMM	1.375	1.440	0.900	1.211
1000	0.75	0.25	DML-IVQR	0.525	1.193	1.200	1.192
1000	0.75	0.10	Oracle-GMM	4.000	3.329	4.000	3.358
1000	0.75	0.10	Full-GMM	3.900	3.362	3.800	3.244
1000	0.75	0.10	DML-IVQR	3.575	3.136	3.600	3.045

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Table A.2: Grid-domain diagnostic: added accepted measure (Continued)

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.005	0.000	0.000
1000	0.90	1.00	DML-IVQR	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	2.050	2.244	2.000	2.050
1000	0.90	0.25	Full-GMM	2.600	2.568	2.200	2.309
1000	0.90	0.25	DML-IVQR	2.000	2.135	2.000	2.125
1000	0.90	0.10	Oracle-GMM	4.000	3.529	4.000	3.530
1000	0.90	0.10	Full-GMM	3.800	3.631	3.825	3.583
1000	0.90	0.10	DML-IVQR	4.000	3.399	4.000	3.353

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table A.3: Grid-domain diagnostic: boundary and full-grid acceptance

n	τ	κ	Estimator	Boundary A_0	Boundary A_1	Boundary A_2	All A_0	All A_1	All A_2
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	0.880	0.770	0.740	0.150	0.070	0.050
1000	0.10	0.25	Full-GMM	0.920	0.840	0.750	0.090	0.030	0.020
1000	0.10	0.25	DML-IVQR	0.950	0.980	0.980	0.240	0.060	0.050
1000	0.10	0.10	Oracle-GMM	1.000	1.000	1.000	0.580	0.430	0.390
1000	0.10	0.10	Full-GMM	0.990	0.980	0.990	0.430	0.210	0.150
1000	0.10	0.10	DML-IVQR	1.000	0.990	0.990	0.560	0.260	0.200
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.400	0.360	0.270	0.070	0.050	0.030
1000	0.25	0.25	Full-GMM	0.560	0.390	0.390	0.060	0.030	0.010
1000	0.25	0.25	DML-IVQR	0.610	0.740	0.740	0.090	0.050	0.020
1000	0.25	0.10	Oracle-GMM	0.940	0.930	0.940	0.470	0.380	0.320
1000	0.25	0.10	Full-GMM	0.960	0.930	0.900	0.320	0.250	0.210
1000	0.25	0.10	DML-IVQR	0.980	0.990	0.990	0.550	0.400	0.290
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.390	0.200	0.150	0.050	0.040	0.020
1000	0.50	0.25	Full-GMM	0.400	0.240	0.260	0.050	0.020	0.020
1000	0.50	0.25	DML-IVQR	0.440	0.340	0.500	0.060	0.020	0.020
1000	0.50	0.10	Oracle-GMM	0.910	0.890	0.830	0.420	0.350	0.320
1000	0.50	0.10	Full-GMM	0.940	0.890	0.850	0.450	0.320	0.280
1000	0.50	0.10	DML-IVQR	0.900	0.930	0.960	0.450	0.360	0.300
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	0.620	0.470	0.370	0.070	0.050	0.040
1000	0.75	0.25	Full-GMM	0.650	0.550	0.530	0.080	0.010	0.000
1000	0.75	0.25	DML-IVQR	0.550	0.450	0.550	0.090	0.060	0.030
1000	0.75	0.10	Oracle-GMM	0.940	0.960	0.930	0.500	0.400	0.380
1000	0.75	0.10	Full-GMM	0.980	0.920	0.920	0.420	0.270	0.220
1000	0.75	0.10	DML-IVQR	0.950	0.970	0.980	0.470	0.360	0.310
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	DML-IVQR	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	0.850	0.760	0.760	0.250	0.090	0.060
1000	0.90	0.25	Full-GMM	0.960	0.800	0.820	0.110	0.060	0.030
1000	0.90	0.25	DML-IVQR	0.820	0.770	0.790	0.210	0.100	0.060
1000	0.90	0.10	Oracle-GMM	0.980	1.000	1.000	0.620	0.450	0.430
1000	0.90	0.10	Full-GMM	1.000	0.990	0.990	0.380	0.110	0.060
1000	0.90	0.10	DML-IVQR	0.990	0.980	0.990	0.510	0.394	0.343

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table A.4: Grid-domain diagnostic: profiles and numerical errors

n	τ	κ	Estimator	R	Succ. A_0	Succ. A_1	Succ. A_2	Fail A_0	Fail A_1	Fail A_2	Stage 1	Stage 2
1000	0.10	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	DML-IVQR	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	DML-IVQR	100	100	99	99	0	1	1	1	0

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table A.5: Grid-domain diagnostic: stabilization within investigated domains

n	τ	κ	**Estimator**	**Investigated-domain status**
1000	0.10	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.10	1.00	Full-GMM	near-stable within investigated domains
1000	0.10	1.00	DML-IVQR	near-stable within investigated domains
1000	0.10	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.25	Full-GMM	not stabilized within investigated domains
1000	0.10	0.25	DML-IVQR	not stabilized within investigated domains
1000	0.10	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.10	Full-GMM	not stabilized within investigated domains
1000	0.10	0.10	DML-IVQR	not stabilized within investigated domains
1000	0.25	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.25	1.00	Full-GMM	near-stable within investigated domains
1000	0.25	1.00	DML-IVQR	near-stable within investigated domains
1000	0.25	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.25	Full-GMM	not stabilized within investigated domains
1000	0.25	0.25	DML-IVQR	not stabilized within investigated domains
1000	0.25	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.10	Full-GMM	not stabilized within investigated domains
1000	0.25	0.10	DML-IVQR	not stabilized within investigated domains
1000	0.50	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.50	1.00	Full-GMM	near-stable within investigated domains
1000	0.50	1.00	DML-IVQR	near-stable within investigated domains
1000	0.50	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.25	Full-GMM	not stabilized within investigated domains
1000	0.50	0.25	DML-IVQR	not stabilized within investigated domains
1000	0.50	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.10	Full-GMM	not stabilized within investigated domains
1000	0.50	0.10	DML-IVQR	not stabilized within investigated domains
1000	0.75	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.75	1.00	Full-GMM	near-stable within investigated domains
1000	0.75	1.00	DML-IVQR	near-stable within investigated domains
1000	0.75	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.25	Full-GMM	not stabilized within investigated domains
1000	0.75	0.25	DML-IVQR	not stabilized within investigated domains
1000	0.75	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.10	Full-GMM	not stabilized within investigated domains
1000	0.75	0.10	DML-IVQR	not stabilized within investigated domains
1000	0.90	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.90	1.00	Full-GMM	near-stable within investigated domains
1000	0.90	1.00	DML-IVQR	near-stable within investigated domains
1000	0.90	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.90	0.25	Full-GMM	not stabilized within investigated domains
1000	0.90	0.25	DML-IVQR	not stabilized within investigated domains
1000	0.90	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.90	0.10	Full-GMM	not stabilized within investigated domains
1000	0.90	0.10	DML-IVQR	not stabilized within investigated domains

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.